STA5001 Bayesian & Computational Statistics

Fall 2026 (First Term)

September 7, 2026 – December 18, 2026

Instructor:	Sheng Jiang	**Time:**	Tuesday, 9:00 AM – 11:50 PM
Email:	jiangsheng@cuhk.edu.cn	**Place:**	CHENGD 207

Description:
This course introduces fundamental Bayesian methodology and computational approaches to Bayesian inference. It begins with commonly used probability distributions, prior distributions, and conjugate Bayesian models. For more sophisticated models and priors, students will learn to implement Monte Carlo, Markov chain Monte Carlo, and variational methods for posterior inference using R and Python.
Applications of Bayesian methods to real-world problems will include Bayesian regression, hierarchical models, latent-variable models, mixture models, and Bayesian nonparametric methods. The course emphasizes statistical foundations, computational implementation, reproducible analysis, and clear communication of statistical findings.

Course Pages:

1. Blackboard.
2. The instructor's personal website for supplementary materials.

Teaching Assistant:

- Jingning Yao, CUHK-SZ email

Tutorials:

- Wednesdays, 18:00pm – 18:50pm, Teaching D Bldg 105

Office Hours:

- Sheng: after each lecture and on Tuesdays from 3:00 PM to 4:00 PM in Dao Yuan 323A, or by appointment.
- Jingning: during tutorials

Prerequisites:
STA2002: Probability and Statistics II, or approval from the academic advisor.

Programming Languages:
We will use R and Python for statistical computing and data analysis. RStudio may be used to write and run R code. Students will also be expected to produce reproducible reports or computational notebooks.

Textbooks:

- Hoff, P. D. (2009). *A First Course in Bayesian Statistical Methods*. Springer.

- Gamerman, D., and Lopes, H. F. (2006). *Markov Chain Monte Carlo: Stochastic Simulation for Bayesian Inference*. Chapman and Hall/CRC.

Useful References:

- Amaral Turkman, A., Paulino, C. D., and Müller, P. (2019). *Computational Bayesian Statistics* (2nd ed.). Cambridge University Press.
- Brooks, S., Gelman, A., Jones, G., and Meng, X.-L. (Eds.). (2011). *Handbook of Markov Chain Monte Carlo*. CRC Press.
- Efron, B., and Hastie, T. (2016). *Computer Age Statistical Inference: Algorithms, Evidence, and Data Science*. Cambridge University Press.
- Murphy, K. P. (2022). *Probabilistic Machine Learning: An Introduction*. MIT Press.

  An electronic copy is available from the book website.
- Murphy, K. P. (2023). *Probabilistic Machine Learning: Advanced Topics*. MIT Press.

  An electronic copy is available from the book website.

Course Project:
A central goal of this course is to develop the ability to conduct a complete Bayesian analysis of real-world data. Through the midterm and final projects, students will work collaboratively to move from a raw or complex dataset to a reproducible statistical analysis.
The project will involve understanding the substantive background, identifying meaningful scientific questions, cleaning and documenting the data, conducting exploratory data analysis, constructing Bayesian models, performing posterior computation, assessing computational and modeling performance, and communicating statistical conclusions.
The midterm project emphasizes substantive background, data preparation, documentation, and exploratory data analysis. The final project emphasizes Bayesian modeling, posterior computation, interpretation, and communication of conclusions.

Project Format and Submission Requirements:
All project reports must use the official NeurIPS 2026 LaTeX template. The template is adopted to provide a consistent professional format; the course-specific requirements below replace the conference submission requirements.

- **Author information:** Reports are not anonymous. The title page must list the full names, student identification numbers, and email addresses of all group members.
- **Formatting:** Students must use the font sizes, margins, line spacing, and single-column layout specified by the NeurIPS 2026 template. Modifying the style file or using formatting commands to circumvent the page limits is not permitted.
- **File format:** Reports must be submitted as PDF files. The submitted PDF must have searchable text, embedded fonts, legible figures, and working references and hyperlinks.
- **Midterm report:** The main text may not exceed four pages. References may appear on additional pages and do not count toward the four-page limit.
- **Final report:** The main text may not exceed nine pages. References may appear on additional pages and do not count toward the nine-page limit.

- **Appendices:** Appendices are optional and may contain supplementary technical details, additional diagnostic results, or additional figures. The main report must be understandable without consulting the appendix. Appendices will be read at the instructor's discretion and must not be used to circumvent the main-text page limit.
- **Required supporting files:** Each group must submit the source code or computational notebooks needed to reproduce the analysis. Groups must also submit the cleaned dataset unless redistribution is prohibited by the applicable license or data-use agreement. If data cannot be redistributed, the group must provide access instructions and code for obtaining or reconstructing the permitted data.
- **Figures and tables:** Figures and tables must be legible at normal viewing size and must have informative captions. All figures and tables must be numbered and discussed in the main text. Axes, units, variables, and relevant uncertainty summaries must be clearly labeled.
- **References:** All data sources, software packages, computational tools, and scholarly sources must be properly cited. References should be formatted consistently.
- **Contribution statement:** Each report must contain a brief contribution statement describing the work completed by every group member.
- **Reproducibility:** The submission should include a README file describing the directory structure, software dependencies, and instructions for reproducing the principal results.
- **NeurIPS checklist:** The NeurIPS paper checklist is not required for this course unless otherwise announced. Students must instead follow the course-specific reproducibility, attribution, and data-ethics requirements in this syllabus.

Possible Sources of Project Data:
Students may propose datasets from any appropriate domain, subject to instructor approval. Possible directions include:

- **Food, menus, and recipes:** Compare regional food traditions, such as Sichuan and Beijing cuisine, using ingredients, flavors, cooking techniques, nutritional information, or recipe networks.
- **Literature and fictional social networks:** Study alliances, conflicts, and their evolution over time in works such as *A Song of Ice and Fire*, *The Lord of the Rings*, or the Four Great Classical Novels of Chinese literature.
- **Text and publication data:** Analyze programs, abstracts, or papers from artificial intelligence and statistics conferences to investigate how research topics emerge, migrate, and evolve.
- **Transportation and network data:** Study traffic flows, mobility patterns, or transportation networks, including data available through scientific-data repositories such as *Scientific Data*.
- **Ocean and environmental data:** Use publicly available oceanographic, climate, or geographic data from sources such as NOAA or C-MAP.
- **Biomedical and population data:** Investigate publicly accessible biomedical datasets or permitted summary statistics from resources related to UK Biobank or similar repositories.

All datasets must be obtained, used, and shared legally and ethically. Groups must not upload confidential, personally identifiable, or access-restricted data to the shared class folder. Controlled-access data, including individual-level UK Biobank data, may be used only in accordance with the applicable authorization and data-use agreement.

Grading Policy:

- **Homework (35%):**
  - Homework will generally be assigned weekly and may include theoretical, computational, and data-analysis problems.
  - Assignments must be submitted electronically through Blackboard.
  - Unless otherwise announced, homework is due at 11:59 p.m. on the stated due date.
  - Late homework will not be accepted.
- **In-class quizzes (15%):**
  - There will be three in-class quizzes, each worth 5% of the final course grade.
  - The quizzes will assess students' individual understanding of recently covered material and provide timely feedback on their learning.
  - Quiz dates are listed in the Important Dates section. Specific coverage will be announced in class and on Blackboard.
- **Midterm project: report and presentation (15%):**
  - Students will work in groups of no more than three.
  - Each group will select a real-world dataset with meaningful complexity and formulate substantive scientific questions that can be studied using the data.
  - Because project topics may come from diverse disciplines, each group must provide sufficient background for a general audience. The group should explain the substantive context, the source of the data, why the problem is interesting, and the scientific questions to be investigated.
  - Each group must clean, organize, and document its dataset. The cleaned dataset, a data dictionary, and relevant documentation must be uploaded to the designated shared Dropbox folder.
  - Each group must submit a PDF report of no more than four pages, excluding references, using the NeurIPS 2026 LaTeX template.
  - The report should describe the substantive background, data source, variables, data-cleaning decisions, exploratory analysis, important patterns in the data, and the scientific questions that will guide the final project.
  - Each group will give a short presentation on Tuesday, November 3, 2026.
  - The midterm report and all required supporting files are due by 11:59 p.m. on Sunday, November 8, 2026.
- **Final project: report and presentation (35%):**
  - Groups will continue working with the dataset and scientific questions developed in the midterm project.
  - The final project will apply suitable Bayesian models and computational methods to answer the scientific questions.
  - Groups should justify their model and prior choices, describe their posterior-inference procedures, evaluate convergence and computational reliability, and assess model fit when appropriate.
  - Each group must submit a PDF report of no more than nine pages, excluding references, using the NeurIPS 2026 LaTeX template.
  - The report should describe the substantive background, data, model, prior choices, computational methods, results, uncertainty quantification, model assessment, interpretation, limitations, and conclusions.
  - Each group will give an oral presentation during the last lecture on Tuesday, December 15, 2026.

  - The final report and all required supporting files are due by 11:59 p.m. on Sunday, December 20, 2026.
  - The final-project grade includes both the written report and the oral presentation.

Presentation Format:

- All presentations must be delivered in person during the assigned class meeting.
- Midterm presentations are limited to five minutes per group, followed by up to two minutes for questions.
- Final presentations are limited to six minutes per group, followed by up to two minutes for questions.
- Presentation slides must be submitted as a PDF file before the beginning of the presentation session.
- All group members must attend the presentation. Each member is expected to participate in the presentation or the question period.
- Presentations must provide enough substantive background for a general statistical audience to understand the dataset, scientific motivation, and importance of the questions being studied.
- Presentations that exceed the allotted time may be stopped to ensure that every group has an equal opportunity to present.

Grading Scale:
Letter grades will be based on students' rankings among all students registered in STA5001. The following table provides the general ranking ranges. Specific letter-grade cutoffs will be determined after considering the overall distribution of final course scores.

Letter Grade	Ranking	Letter Grade	Ranking
A	Top 20%	A−	21%–40%
B+, B, B−	41%–80%	C, D, F	81%–100%

The precise allocation of B+, B, and B− grades within the 41%–80% range, and of C, D, and F grades within the 81%–100% range, will be determined after reviewing the final score distribution.

Important Dates:

First Term Begins	Monday, September 7, 2026
First STA5001 Lecture	Tuesday, September 8, 2026
Add/Drop Period	September 6 – September 18, 2026
Quiz 1	Tuesday, September 22, 2026
Mid-Autumn Festival	September 25 – September 27, 2026
National Day Holiday	October 1 – October 7, 2026
No Class	Tuesday, October 6, 2026
Quiz 2	Tuesday, October 27, 2026
Midterm Project Presentations	Tuesday, November 3, 2026
Midterm Project Report Due	Sunday, November 8, 2026, 11:59 p.m.
Quiz 3	Tuesday, December 1, 2026
Final Project Presentations	Tuesday, December 15, 2026
First Term Ends	Friday, December 18, 2026
Final Project Report Due	Sunday, December 20, 2026, 11:59 p.m.
Course Examination Period	December 20 – December 27, 2026
Independent Exploration Period	December 28, 2026 – January 10, 2027

Course Policies:

- We shall strictly follow the **honor code** and the university's academic misconduct policy.
- Feel free to interrupt me and ask questions during class.
- Regular attendance is essential for success in this course and is expected. In-class quizzes and project presentations require attendance.
- Each student is expected to contribute substantively to all group-project work. Each project report must include a brief contribution statement describing the work completed by every group member.
- All analyses must be reproducible. Groups should retain and submit their code, notebooks, data documentation, and other supporting files when requested.
- English is my second language. If you spot any errors or have suggestions, please let me know.